

Agentic Artificial Intelligence

Elective Module

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Your Lecturer



MARS



ResearchGate



LinkedIn

- **Full professor** of AI & Data Management (since 2009)
- **Adjunct Professor**, Carleton University, Canada
- **Partner** und AI Practice Lead, [4-advice GmbH](#)
- **Research**
 - Multi-agent systems
 - LLM-based agents, generative agents, and Agentic AI
 - Agent-based modeling and simulation
 - Cooperation in multi-agent systems
 - Multi-agent (deep) reinforcement learning (MA(D)RL)
 - Digital twins
 - Geoinformatics
- **Background**
 - 10 yrs in consulting, sales, and management
 - PhD in CS (Univ. of Kiel)
 - Diploma in CS (TU Dortmund)

The MARS Group



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- Decision-support systems
- AI-based cooperation
- AI in earth observation & sustainability
- Human and animal behavior modeling
- Transportation and smart cities
- Logistics
- Spread of infectious diseases
- Citizen science
- Coordination of drone swarms
- Serious games
- ...



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Orientation

- Goals
 - Have fun!
 - Experimentation
 - Interdisciplinary research
 - Applied computer science
 - Intro to Agentic AI
 - Preparation for Bachelor's Thesis (and Master)
- Method of examination
 - I. Pre-requisite for examination ("PVL")
Active participation in lab projects + final presentation
 - II. Term paper
Topic selection until **July 1st, 2026** (end of day)
Submission deadline (PDF, in Moodle): **August 16th, 2026** (end of day)

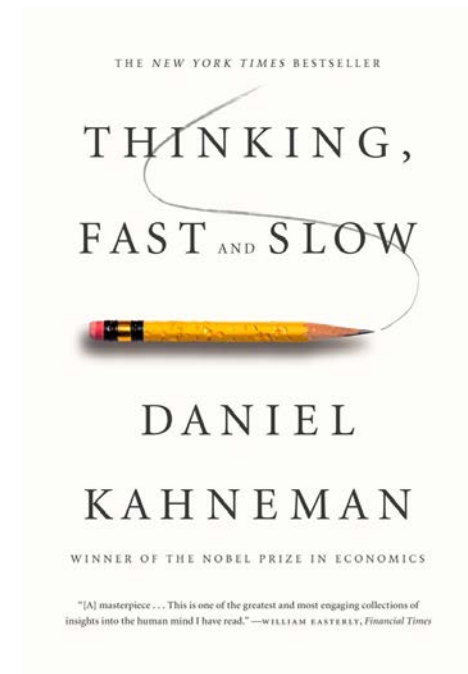
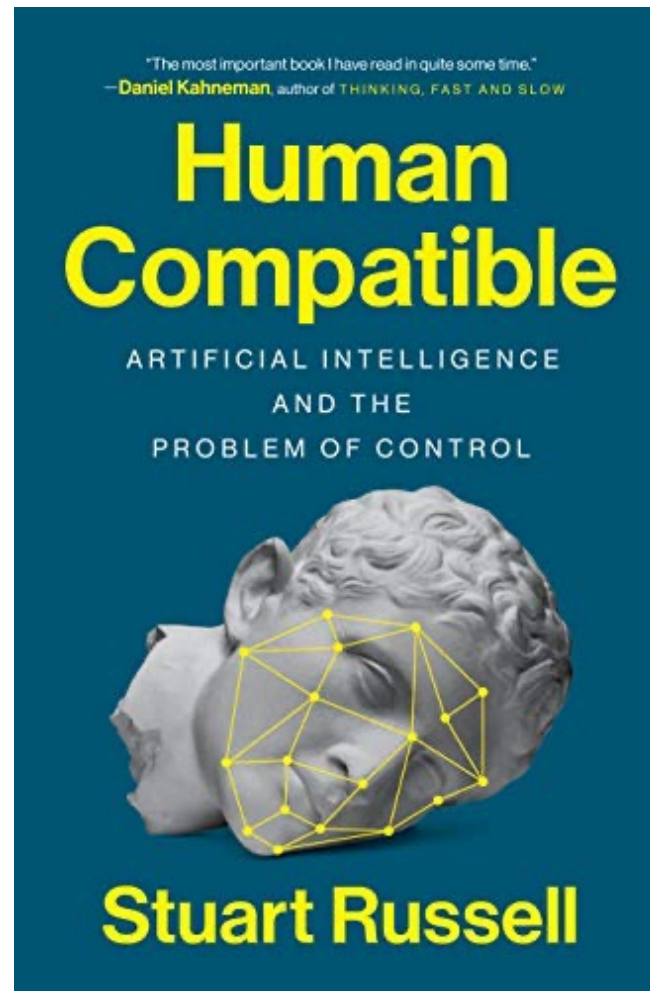
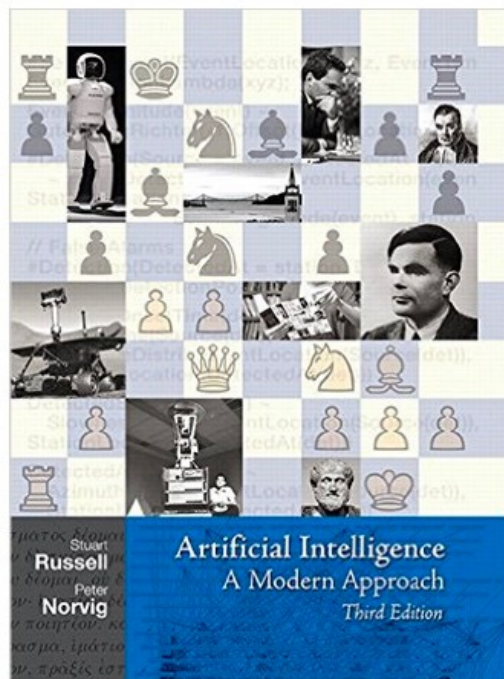
Org stuff

- Each **learning unit** consists of short lecture parts, links to YouTube videos, and web pages as well as technical/scientific papers.
- **Theoretical parts** and **practical exercises** alternate in each lecture. Therefore, please bring a laptop to the lecture hall.
- During the labs, you will work in groups of 3-5 students on a selected **term project**.
- Fun fact: Most of the slides used in this lecture were created using Anthropic's Claude Opus 4.6 and Sonnet 4.6.

Schedule

Week	Week	Week	Week	Deadlines
16	14.04.26	Lec01	LabG01.01	
17	21.04.26		LabG02/LabG03.01	
18	28.04.26	Lec02	LabG01.02	
19	05.05.26		LabG02/LabG03.02	
20	12.05.26	Lec03	LabG01.03	
21	19.05.26		LabG02/LabG03.03	
22	26.05.26	Lec04	LabG01.04	
23	02.06.26		LabG02/LabG03.04	
24	09.06.26	Lec05	LabG01.05	
25	16.06.26		LabG02/LabG03.05	
26	23.06.26	Lec06	LabG01.06	
27	30.06.26		LabG02/LabG03.06	Deadline Topic Selection (1.7.)
28	07.07.26	Lec07	LabG01.07	
29	14.07.26		LabG02/LabG03.07	
	16.08.26			Deadline Term Paper

Recommended Readings



Communications

- Moodle: <https://moodle.haw-hamburg.de/course/view.php?id=10002>
Self enrolment key: ai2026



- MS Teams
Teamcode: yhlz42u

The Rationale behind this Course

What is an agent? →

How does agents perceive their world, decide, and learn? →

How does agents become capable of utilizing tools and data? →

How does agents cooperate? →

How can agentic AI systems be designed and implemented? →

How can agent-based AI systems be designed securely?

Neighbourhood Chat

Your expectations and comments (5 min)

